

INTRODUCTION TO INFORMATION & COMMUNICATION TECHNOLOGIES (ICT)

Course Lecture Notes

TOPIC

Introduction to Computers

SUB-TOPIC: History and Evolution of Computers

Item	Details
Subject	Introduction to Information & Communication Technologies (ICT)
Unit / Chapter	Introduction to Computers
Topic of the Day	History and Evolution of Computers
Level	DIT PART-I
Prepared by	Hafiz Muhammad Usman (Lecturer, Computer Science)

1. Learning Objectives

By the end of this lecture, students will be able to:

- Define what a computer is and identify its fundamental characteristics.
- Trace the historical evolution of computing devices from ancient calculating tools to modern computers.
- Classify the generations of computers and describe the defining technology of each generation.
- Compare the generations of computers on the basis of technology, speed, size, and cost.
- Identify key inventors, machines, and milestones that shaped modern computing.
- Appreciate the trajectory of computing evolution and relate it to present-day and future computing trends.

2. What Is a Computer?

Definition: Computer

An electronic device that accepts data as input, processes that data according to a set of instructions (a program) stored in its memory, and produces meaningful information as output — while also being capable of storing data for future use.

A computer is best understood through its functional cycle, often called the IPO Cycle (Input–Process–Output), extended with Storage:

Stage	Function	Example
Input	Data and instructions are entered into the computer.	Typing text on a keyboard, scanning a barcode
Process	The CPU manipulates and processes the raw data according to instructions.	Calculating the total of a shopping bill
Output	The processed data (information) is presented to the user.	Displaying the bill on a monitor, printing a receipt
Storage	Data, instructions, and results are saved for future use.	Saving a file to a hard disk or USB drive

2.1 Key Characteristics of a Computer

Characteristic	Explanation
Speed	Performs millions to billions of calculations per second (measured in MIPS/FLOPS).
Accuracy	Produces error-free results, provided the input data and instructions are correct (GIGO principle — Garbage In, Garbage Out).
Diligence	Can perform repetitive tasks continuously without fatigue, boredom, or loss of accuracy.

Characteristic	Explanation
Storage Capacity	Can store enormous amounts of data and retrieve it instantly when required.
Versatility	Can perform a wide variety of tasks — from calculations to multimedia to communication.
Automation	Once a program is loaded, it can execute instructions automatically with minimal human intervention.
No Intelligence (Feeling)	A computer has no intelligence of its own — it can only do what it is instructed (programmed) to do.

3. History and Evolution of Computers

The history of computers is a continuum of human effort to reduce the time and effort spent on calculation and data management. It can be studied in two broad phases:

- **Pre-Mechanical / Early Era** — manual and mechanical calculating devices (before electronics).
- **Modern / Electronic Era** — classified into five "Generations of Computers" based on the core electronic technology used.

3.1 Early Calculating Devices (Before Electronic Computers)

Device / Era	Approx. Period	Description
Abacus	~2400 BC (Babylon/China)	Considered the earliest known calculating device; used beads on rods to perform addition and subtraction.
Napier's Bones	1617 (John Napier)	A manual calculating tool using numbered rods to perform multiplication and division through addition.
Pascaline	1642 (Blaise Pascal)	The first mechanical calculator, capable of addition and subtraction using gears and wheels.
Stepped Reckoner	1673 (Gottfried Leibniz)	Improved Pascal's design; could perform multiplication and division as well.
Jacquard's Loom	1801 (Joseph Marie Jacquard)	Used punched cards to control weaving patterns — introduced the concept of programmable machines and punched-card input.
Difference Engine	1822 (Charles Babbage)	A mechanical device designed to compute and print mathematical tables automatically.
Analytical Engine	1837 (Charles Babbage)	A theoretical general-purpose mechanical computer with memory, a control unit, and input/output — never fully built, but conceptually the ancestor of the modern computer.
Tabulating Machine	1890 (Herman Hollerith)	Used punched cards to tabulate the US Census data; Hollerith's company later became part of IBM.

✔ Why This Matters

- Charles Babbage is called the "Father of the Computer" because the Analytical Engine had all the essential parts of a modern computer: input, memory (store), a processing unit (mill), and output.

Device / Era	Approx. Period	Description
• Ada Lovelace, who worked with Babbage, wrote the first algorithm intended for the Analytical Engine and is regarded as the world's first computer programmer.		

3.2 The Five Generations of Computers

Modern computers are classified into five generations, primarily based on the core electronic technology used to build them.

Generation 1: First Generation (1940 – 1956) — Vacuum Tubes

Definition: Vacuum Tube

A glass tube that controls electric current flow and was used as the primary switching and amplifying component in early electronic computers.

Aspect	Details
Core Technology	Vacuum tubes and magnetic drums for memory
Key Machines	ENIAC (1946), UNIVAC I (1951), EDVAC, Mark I
Programming Language	Machine language (binary code, 0s and 1s)
Size	Extremely large — occupied entire rooms
Speed	Very slow (milliseconds)
Heat & Power	Generated enormous heat; consumed huge electricity; required special cooling
Reliability	Low — vacuum tubes burned out frequently
Input/Output	Punched cards and paper tape
Example • ENIAC (Electronic Numerical Integrator and Computer) weighed about 30 tons, occupied 1,800 square feet, and used around 18,000 vacuum tubes.	

Generation 2: Second Generation (1956 – 1963) — Transistors

Definition: Transistor

A small semiconductor device that replaced vacuum tubes to amplify and switch electronic signals — smaller, faster, cheaper, and more reliable.

Aspect	Details
Core Technology	Transistors (invented at Bell Labs, 1947) replacing vacuum tubes
Key Machines	IBM 1401, IBM 7090, UNIVAC 1108
Programming Language	Assembly language, and early high-level languages (COBOL, FORTRAN)

Aspect	Details
Size	Smaller than first generation, but still large
Speed	Faster than vacuum tube computers (microseconds)
Heat & Power	Much less heat and power consumption
Reliability	Improved significantly
Input/Output	Punched cards, magnetic tape

Generation 3: Third Generation (1964 – 1971) — Integrated Circuits (ICs)

Definition: *Integrated Circuit (IC)*

A small chip (silicon wafer) that contains many transistors and electronic components combined together to perform the functions of a large circuit.

Aspect	Details
Core Technology	Integrated Circuits (ICs), also called silicon chips
Key Machines	IBM System/360, PDP-8, PDP-11
Programming Language	High-level languages (FORTRAN, BASIC, COBOL); Operating Systems introduced
Size	Considerably smaller — desk-sized
Speed	Faster (nanoseconds); multitasking became possible
Heat & Power	Further reduced due to compact circuitry
Reliability	High
Input/Output	Keyboards and monitors introduced; Operating Systems allowed a device to run many applications at once

Generation 4: Fourth Generation (1971 – 2010) — Microprocessors

Definition: *Microprocessor*

A single silicon chip that contains an entire Central Processing Unit (CPU) — thousands to billions of transistors integrated on one chip.

Aspect	Details
Core Technology	Microprocessor (Intel 4004, 1971 — first commercial microprocessor)
Key Machines	Apple II, IBM PC (1981), Macintosh (1984)
Programming Language	C, C++, Java; Graphical User Interfaces (GUIs) emerge
Size	Personal computers — desktop and, later, laptop-sized
Speed	Extremely fast (picoseconds); multi-core processors emerge
Heat & Power	Low power consumption; energy-efficient

Aspect	Details
Reliability	Very high
Input/Output	Mouse, GUI, keyboard; networking and the Internet emerge
✔Landmark Fourth-Generation Milestones • 1971 — Intel 4004 microprocessor launched. • 1981 — IBM Personal Computer (PC) introduced, standardizing personal computing. • 1984 — Apple Macintosh popularized the Graphical User Interface (GUI) and mouse. • 1990s onward — Rise of the Internet, connecting computers worldwide.	

Generation 5: Fifth Generation (2010 – Present and Beyond) — Artificial Intelligence

Definition: Artificial Intelligence (AI) The branch of computing focused on building systems that can perform tasks which typically require human intelligence, such as learning, reasoning, problem-solving, and understanding natural language.

Aspect	Details
Core Technology	AI, Machine Learning, Natural Language Processing, Quantum Computing (emerging), parallel processing
Key Examples	Smartphones, voice assistants (Siri, Alexa), self-driving cars, cloud computing, generative AI (ChatGPT, Claude)
Programming Language	Python, high-level AI frameworks (TensorFlow, PyTorch)
Size	Highly compact — wearable and embedded devices; also large-scale cloud data centers
Speed	Enormous parallel processing power; GPUs/TPUs accelerate AI workloads
Reliability	Very high, with self-correcting and adaptive systems
Input/Output	Voice, touch, gesture, biometric input; AI-generated multimedia output

4. Comparative Summary of All Five Generations

Generation	Period	Technology Used	Speed	Size	Example
First	1940–1956	Vacuum Tubes	Milliseconds	Room-sized	ENIAC, UNIVAC I
Second	1956–1963	Transistors	Microseconds	Large cabinet	IBM 1401
Third	1964–1971	Integrated Circuits	Nanoseconds	Desk-sized	IBM System/360
Fourth	1971–2010	Microprocessors	Picoseconds	Desktop/Laptop	IBM PC, Macintosh
Fifth	2010–present	AI & Parallel Processing	Extremely high (parallel)	Compact/Cloud-based	Smartphones, AI assistants

5. Quick-Reference Timeline of Milestones

Year	Milestone
~2400 BC	Abacus used in Babylon for calculations
1642	Blaise Pascal builds the Pascaline mechanical calculator
1837	Charles Babbage designs the Analytical Engine
1843	Ada Lovelace writes the first algorithm for a machine
1946	ENIAC — first general-purpose electronic computer
1947	Transistor invented at Bell Labs
1958	First Integrated Circuit developed
1971	Intel 4004 — first commercial microprocessor
1981	IBM Personal Computer launched
1984	Apple Macintosh introduces GUI to the mass market
1991	World Wide Web made publicly available
2007	Smartphones (iPhone) redefine personal computing
2020s	Generative AI and large language models become mainstream

6. Key Terms & Glossary

Term	Meaning
Hardware	The physical, tangible components of a computer (e.g., CPU, monitor, keyboard).
Software	The set of instructions (programs) that tell hardware what to do.
CPU (Central Processing Unit)	The "brain" of the computer that executes instructions and processes data.
Generation	A stage in computer history distinguished by a major shift in the core technology used to build computers.
Moore's Law	An observation by Gordon Moore (1965) that the number of transistors on a microchip doubles approximately every two years, while cost decreases.
GUI (Graphical User Interface)	A visual way of interacting with a computer using icons, windows, and a mouse, rather than typed text commands.

7. Summary

✔ Key Takeaways

- A computer processes data into information through the Input–Process–Output–Storage cycle.
- Computing evolved from manual tools (Abacus) to mechanical calculators (Pascaline, Analytical Engine) and finally to electronic computers.

- Computers are classified into five generations: Vacuum Tubes → Transistors → Integrated Circuits → Microprocessors → Artificial Intelligence.
- Each generation brought improvements in speed, size, cost, reliability, and power efficiency.
- Charles Babbage is the "Father of the Computer"; Ada Lovelace is considered the first programmer.
- The trend across generations shows continuous miniaturization, increasing speed, and decreasing cost — a trend expected to continue with quantum and AI-driven computing.

8. Review Questions (For Practice / Assignment)

- Define a computer and explain the Input–Process–Output–Storage cycle with a real-life example.
- Explain why Charles Babbage is regarded as the "Father of the Computer."
- Differentiate between the first and second generations of computers on the basis of technology used.
- Construct a table comparing all five generations of computers on any four parameters of your choice.
- Discuss the role of the microprocessor in transforming computers into personal, everyday devices.
- In your opinion, what technological shift will define the "Sixth Generation" of computers? Justify your answer.